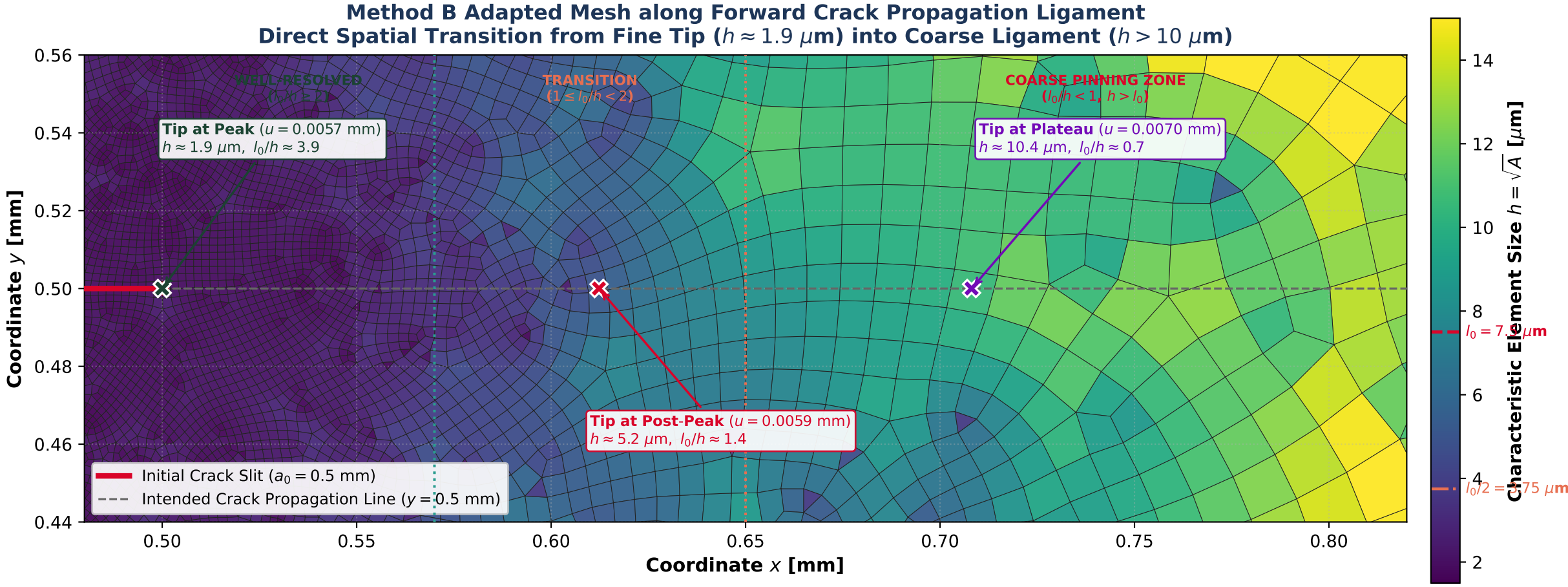

Method B Adapted Mesh along Forward Crack Propagation Ligament
Direct Spatial Transition from Fine Tip ($h \approx 1.9\ \mu$m) into Coarse Ligament ($h > 10\ \mu$m)
WELL-RESOLVED
($l_0/h \geq 2$)
TRANSITION
($1 \leq l_0/h < 2$)
COARSE PINNING ZONE
($l_0/h < 1$, $h > l_0$)
Tip at Peak ($u = 0.0057$ mm)
$h \approx 1.9\ \mu$m, $l_0/h \approx 3.9$
Tip at Plateau ($u = 0.0070$ mm)
$h \approx 10.4\ \mu$m, $l_0/h \approx 0.7$
Tip at Post-Peak ($u = 0.0059$ mm)
$h \approx 5.2\ \mu$m, $l_0/h \approx 1.4$
Initial Crack Slit ($a_0 = 0.5$ mm)
Intended Crack Propagation Line ($y = 0.5$ mm)
Coordinate y [mm]
Coordinate x [mm]
Characteristic Element Size $h = \sqrt{A}$ [μm]
$l_0 = 7.5\ \mu$m
$l_0/2 = 3.75\ \mu$m
0.56
0.54
0.52
0.50
0.48
0.46
0.44
0.50
0.55
0.60
0.65
0.70
0.75
0.80
14
12
10
8
6
4
2